

Design Thinking: A New Foundational Science for Engineering*

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Design thinking has seen rapid growth since mid-2000s far beyond the engineering and arts disciplines traditionally concerned with design. In universities, it is increasingly being used to create programs where graduates from multiple disciplines can learn to develop a design orientation to problem solving. This rising popularity of design thinking as a generalist training methodology positioned alongside the MBA program while helpful, obscures the central role it plays in the engineering disciplines. In this paper, we argue for design thinking to be recognized as a foundational science for engineering alongside Physics, Chemistry and Biology.

Keywords: design thinking measurement; design teams; product based learning; technology innovation; innovation eco-systems; economic growth

1. Introduction—What are the criteria for becoming a foundational science for engineering?

In his State of the School address in May 20, 2003, the dean of engineering at Stanford University, Jim Plummer, announced the launch of a new initiative, the creation of a bioengineering department. 24 Incremental faculty billets had been approved by the provost to be shared between the school of Engineering and the school of medicine. The dean argued that our ability to measure biological systems at the molecular level and to understand the molecular basis of living systems had turned biology from being a qualitative science to being a quantitative science like physics, chemistry, and mathematics. He concluded saying Biology is ready to become a new foundational science for engineering. Could the same be said of design thinking? In this paper we will make the case that design thinking, is not only foundational to engineering, it lies at the core of what engineers know and do. We will further argue that while in the past, Engineers have looked to other fields to validate their work, Engineering as a discipline has now come of age, and it is time to acknowledge the critical role design plays in engineering [1]. Finally, we will show that design thinking can indeed be measured, and by using the same criteria as was used in Biology, it will be possible to conclude that design thinking is ready to become a new foundational science of engineering.

2. Context—Is context critical in design thinking?

Unlike the natural sciences, which strive for context

independence [2, 3], design thinking and indeed engineering is highly context dependent. Therefore in this paper, we will describe our experiences at a very local and specific level. This will include references to Stanford University, IDEO a product consulting firm in Palo Alto, the Silicon Valley, the United States of America, and reference to specific people working in these environments. This is not in any way to play down the contributions of people working in other regions of the US, and there are several people that come to mind, however the contextual nature of the phenomenon we are studying and practicing, requires us to make this choice. Indeed we make the following functional comparison:

“The Silicon Valley is to the Information Revolution what the Steam Engine was to the Industrial Revolution.”

Thus just like all the components of the Steam Engine like the Boiler and Pumps have to be co-located and connected for energy to be generated, the design components of an engineering and economic engine like the Silicon Valley have to be co-located and connected. To do otherwise is meaningless from an engineering point of view. Once built and reasonably understood, the Steam Engine can be rebuilt and improved upon in other contexts, because then, the initial and boundary conditions can be well accounted for. Therefore we extend our sincere apologies to those who may feel their work and contributions have not been highlighted in this paper. We hope in future, we will have a way of adding multiple and redundant citations, if anything to highlight similar and related work going on in other geographies. With that caveat in place, we will now describe the context

of our work, and our primary motive is to make scientific work such as corroboration and replication possible.

Engineering has been often been described as the application of science and mathematics to the needs of society, however, up until now, we knew and taught our students very little about finding out and understanding the needs of society. Thus we indirectly turned out half-baked engineers, engineers who could only serve the needs of the military, and engineers who were easily exploited by others. With the advent of the design-thinking paradigm, this situation has changed. Engineers are now taught how to engage with society through the use of empathy, multiple points of view, teamwork, prototyping, and iterative testing and feedback. In effect, a critical link has been made between engineering analysis, the motion, space and materials science facing part of the discipline and engineering design, the human facing part of the discipline.

As engineering faculty, most of us did not teach engineering design either because we did not understand it and/or it included humans in the loop, which not unlike the phenomenon of friction and inertia, made the formulation of problems more difficult not to mention the derivation of solutions. This attitude, due in part to a lack of understanding and in part to fear, often made some of us have some contempt towards the broader discipline of design. In reality we have often done a limited amount of design especially for problems where we can conveniently ignore the unpredictable human element. Based on the more recent recognition of critical role played by design in the profitability of companies, the cumulative work of the engineering design research community and recent breakthrough in our ability to measure design behavior, we can now change this situation for the better.

This breakthrough at Stanford cannot be divorced from the location of Stanford within the Silicon Valley and the role the University has played in impacting society. Wherein the dean of engineering, Frederick Terman, himself an engineer, reconceptualized the role of the university as follows and we quote:

“Universities are rapidly developing into more than mere places of learning. They are becoming major

economic influences in the nation’s industrial life, affecting the location of industry, population growth, and the character of communities. Universities are in brief a natural resource just as are raw materials, transportation, climate, etc.” [4].

Stanford has thus led the way in the accelerated creation of new ventures, new markets, and increased rate of capital formation. [5]. The foregoing really begs the question—just what is design thinking? And what if any is its relationship to engineering?

3. Definition—What is design thinking?

In arriving at an understanding of design thinking or design more generally, we have found it useful to sketch out the five Venn diagrams shown in Fig. 1, and ask people to identify which most closely matches their understanding of design. Reading from left to right we hope to evoke the following questions—Is design synonymous with engineering?; Is design at the core of engineering?; Is engineering one of the sub-fields of design?; Is design a partially separate and distinct field from engineering?; and finally—Is design a completely separate field from engineering? So we encourage our readers to take a moment to make their selection at this point before reading further.

The term design thinking was used as early as 1991. It was the title of the Proceedings of a Workshop meeting titled “Research in design thinking,” held at the Faculty of Industrial Design Engineering, Delft University of Technology, from May 29–31. However the current popularity of the term can be attributed to the product development consultancy IDEO and its founder David Kelley. It is useful here to get a few snapshots of how Kelley sees and defines design.

“What led you to find IDEO?” “Well, I had just fallen in love with design at Stanford. I was a regular engineer before I got to Stanford where I got into a program that taught the human side of engineering. I really fell in love with the notion of doing something for humans rather than just doing something technologically. Silicon Valley was just starting to do really well and then it became clear that a lot of companies had the need to have somebody who was sensitive to the design, the aesthetics, the humans and the engineering stuff. That

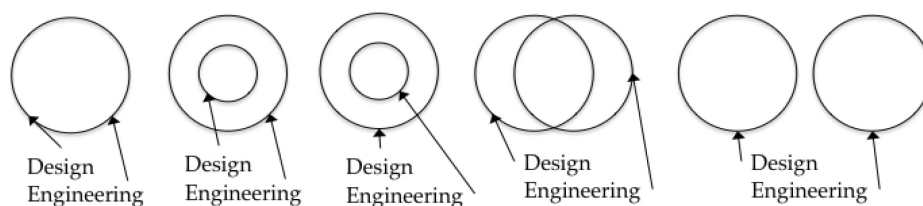


Fig. 1. Defining design relative to engineering.

combination seemed really obviously needed in Silicon Valley. That made it easier to start IDEO” [6].

“What I mean by design is doing things with intention, trying to decide what’s important to somebody, building a bunch of prototypes and showing them around, developing a point of view and getting it out so that it has impact in the world. So design is really a process of making impact on the world by doing this kind of creation of something new to the world and then getting it out there” [6].

“You’re sitting here today because we moved from thinking of ourselves as designers to thinking of ourselves as design thinkers,” . . . “What we, as design thinkers, have, is this creative confidence that, when given a difficult problem, we have a methodology that enables us to come up with a solution that nobody has before” [7].

“Design thinking is a human-centered approach to innovation that draws from the designer’s toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success. Thus, the method focuses on the three main elements of a product or solution: people, technology, and business, all of which evolve around the customer” [8].

We have used several quotations from Kelley, because it highlights a phenomenon we have found to be prevalent among design thinkers—namely a tolerance for ambiguity [9]. Design thinkers on the other hand tell us, they observe things from multiple perspectives. Perhaps the three most important features in the foregoing quotations are that (1) there is no one single definition of design thinking and an important part of the process is “developing a point of view;” (2) the process of design is emergent, and non deterministic; (3) as Kelley points out there is a coherent, integrated, and concerted effort towards creating something new to the world, for the customer, and getting it out there.

This view of design, in a way, attempts to overcome the problem of coherence that has dogged the social sciences. In 1998, E. Wilson made a thought provoking observation in comparing the rate of progress in the medical vs. the social sciences. He remarked:

“There is also progress in the social sciences, but it is much slower, and not at all animated by the same information flow and optimistic spirit . . . The crucial difference between the two domains is consilience: The medical sciences have it and the social sciences do not. Medical scientists build upon a coherent foundation of molecular and cell biology. They pursue elements of health and illness all the way down to the level of Biophysical Chemistry . . . Social scientists by and large spurn the idea of the hierarchical ordering of knowledge that unites and drives the natural sciences. Split into independent cadres, they stress precision in words within their specialty but seldom speak the same technical language from one specialty to the next” [10].

We have been aware of this problem in the social sciences for a while, and it can be a stumbling block

to progress in design thinking. To guard against such a situation while advancing the field, we have found the definition used by Herbert Simon in his book—*The Sciences of the Artificial*—to be the most parsimonious and comprehensive. The parsimony principle is basic to all science and tells us to choose the simplest scientific explanation that fits the evidence. In the next paragraph, we present Simon’s definition of design:

“Historically and traditionally, it has been the task of the science disciplines to teach about natural things: how they are and how they work. It has been the task of engineering schools to teach about artificial things: how to make artifacts that have desired properties and how to design. Engineers are not the only professional designers. Everyone designs who devises courses of action aimed at changing existing situations into preferred ones. The intellectual activity that produces material artifacts is no different fundamentally from the one that prescribes remedies for a sick patient or one that devises a new sales plan for a company or a social welfare policy for a state. Design, so construed, is the core of all professional training; it is the principal mark that distinguishes the professions from the sciences” [11].

In this way, the first three Venn diagrams will satisfy this definition of design, which is the one we have adopted for our research. We also hope the reader will see the similarity between Kelley, the practitioner’s view and Simon, the researcher’s view. Having defined design, the natural question that follows is whether it can be improved, and this will be covered in the next section.

4. Can the process of design be improved upon in a systematic way?—previous approaches

There have been several lines of work that have been aimed at improving design in a systematic way. We will mention four approaches here to give a broader view of the research enterprise whose collective and cumulative effort are only now coming to a head. These remain to this day, critical and important milestones.

4.1 Approach #1: augmenting human intellect [12]

The project on augmenting human intellect was championed by Doug Engelbart and his team at the then Stanford Research Institute, SRI. This was the precursor to the Mouse, Windows, Hyperlinks, and the World Wide Web. Doug, an electrical engineer, felt that the rate at which problems in the world were growing was much faster than the rate at which solutions were being developed. Therefore, they set about looking for ways to use other parts of the human body for problem solving. The idea of the mouse for example uses our hands.

The links we have on the web pages mimics the way we recall things using our associative memory. We can jump from one idea to another idea while reimagining and recreating the context in the process. There were several other inventions made by this group in the 1960's several of which have not been inserted into the public space.

4.2 Approach #2: automating human intelligence [13]

Allan Turing is often called the father of artificial intelligence, and his exploits during the Second World War are legendary. Then there was the work by Herbert Simon and Allen Newell and several others at Carnegie Mellon University focused specifically on Problem Solving [14] and Design. This line of work can be considered the precursor to several firsts in artificial intelligence such as the IBM chess player named deep blue and later Jeopardy quiz show game winner the IBM Watson computer.

4.3 Approach #3: situating human cognition, learning and action

We argued earlier that Design is context dependent and over the years several social scientists and computer scientists have been able to describe the different ways in which the context shapes outcome of the activity. We recall here the work of the following researchers. Note that unlike hardware like the computer mouse, and interactive algorithms like deep blue and Watson, social scientists write books, it is their design artifact. So in Table 1 we have provided a cursory list of works emphasizing the important role played by context.

Thus we can see, that a lot has been written, and a lot has already been built; yet we do not seem to have a systematic way to improve design. For our purposes, we would like to improve design in a way specific to an Engineering viewpoint. And here we encountered in the above approaches, three main limitations, which will be discussed in the next section.

5. Limitations—What are the current constraints to improving design in a systematic manner?

The three limitations are (1) Engineering-Design is a very broad and dynamic field and coverage is difficult; (2) Nobody knew what performance improvement really looked like; (3) There were no standard ways of measuring performance. We will briefly discuss each in turn.

5.1 Limitation #1 engineering design is a very broad field

Engineering Design is quite a broad discipline and practiced in a variety of industrial contexts. For example, the defense sector, the health sector, the automobile sector, the housing sector, the transportation sector, the communication sector, the energy sector, and we could go on and on. Medicine also has several sectors—Orthopedic, Ophthalmology, Psychiatry, Obstetrics and Gynecology, Urology, Neurology, Internal Medicine, and so on. However there is one difference. In medicine it all revolves around the tangible human body. In engineering, the nature of the client's problem is different, the media is different, the laws and regulations are different, the scale of the artifacts are different, to name a few. There isn't a unifying whole.

5.2 Limitation #2. Nobody really knew what performance looked like

We are reminded here of a story told by Doug Engelbart to explain to people the function of the mouse. In the 1960s when it was invented, people had a hard time figuring out why it was necessary. Engelbart, we assume out of some frustration, fastened a heavy mud brick to a pencil and had people write with this weighted pencil, and then removed the brick and had them write again, so they could experience the difference.

Well the same is true about engineering design. Economics as a discipline measures factory output—inventories and production rates. They

Table 1. A Sample of the Research describing the relationship between context and human performance

- 1 Vygotksy, L. (1934/1962). *Thought and Language* [15].
- 2 Dewey, J. (1938). *Experience and Education* [16].
- 3 Bandura, A. (1977). *Social Learning Theory* [17].
- 4 Suchman, L. (1987). *Plans and Situated Actions: The Problem of Human/Machine Communication* [18].
- 5 Brown, J. S., Collins, A., Duguid, P. (1989). *Situated cognition and the culture of learning* [19].
- 6 Lave, J., Wenger, E. (1990). *Situated Learning: Legitimate Peripheral Participation* [20].
- 7 Donald, M., (1991) *Origins of the modern mind: Three stages in the evolution of culture and cognition*. Cambridge, MA: Harvard University Press [21].
- 8 Clancey, W. J. (1997). *Situated Cognition: On Human Knowledge and Computer Representations* [22].

measure nothing related to the intellectual activity we call design.

5.3 Limitation #3. Lack of Standard Measurement Units

In traditional engineering and the natural sciences, we typically measure variations in terms of seven basic units, the SI units. These are—Mass, Length, Time, Temperature, Current, Light Intensity, and Quantity of Substance. All other constructs can be built from these primitives. So for example the construct of “force” is composed of—mass, length, and time. Thus in the absence of such standards, the coordination of progress in design research and practice will be very arduous.

6. Overcoming Limitations—How can the constraints to improving design be redesigned and minimized?

Luckily we were able to find resources that allowed us to overcome the first two limitations.

6.1 Overcoming limitation #1 engineering design is a very broad field

At Stanford, we had been running an Industry sponsored design course, ME210 (presently called ME310) since 1970. In this course a wide variety of companies pay the cost of materials and services for teams of graduate student to work on real industry problems [23]. Today the list of companies includes those in the automobile, health care, software, defense, and several others. Thus, when the course is in session, we could observe in one place, teams of engineers working on problems across a wide spectrum of industries, and we could interact with them. The logistics of such a wide coverage could be reduced if we used the class as our laboratory, which is what we did.

6.2 Overcoming limitation #2. Nobody really knew what performance looked like.

Here we were very fortunate. It was the 80 and 90s, and Toyota had become the bestselling car in America. This feat was perceived as a national security threat and led to several important changes in industry, academia, and government in the US [24–26]. In the national science foundation, the design theory and methodology field was created and charged as follows:

“... the results of research in this field can improve the creativity of engineering design and the effectiveness of product design and development, leading to improved national productivity and international competitiveness” [27].

There were a lot of researchers studying the internal

processes of Japanese companies, and we will describe here the results of two such studies.

Study 1: In a study of the number of patents filed by US, European, and Japanese companies from 1968 to 1986, US companies filed 2500 in 1968 and this gradually fell to 1500 by 1986. The European companies filed approximately 800 patents in 1968 and this rose to about 1400 in 1986. The Japanese companies filed below 100 in 1968 and this rose to about 2200 in 1986, a jump of over 2000 in 15 years, surpassing those of the US and Europe by almost 1000 [28].

In another study, Kim Clark and Takahiro Fujimoto spent 5 years studying automobile companies in Japan, Europe, and the US. They were able to illustrate that the Japanese were able to produce better cars in shorter lead times because their design and manufacturing teams shared information earlier in the process and more frequently [29].

They called this technique, overlapping problem solving. This technique showed the interrelationship between three variables—knowledge level, communication frequency, and lead-time. The technique later became known as concurrent engineering. This was the first large-scale effort to directly link knowledge work to economic impact. Now we had a vague idea what performance could look like and how important multi-disciplinary teamwork was to engineering. This left the third problem of finding standard ways of measuring performance.

6.3 Overcoming Limitation #3. Lack of standard measurement units

We were not able to find any way around this limitation. We therefore created a laboratory to study the class. In our lab we focused on developing metrics of the design process, ways to measure the process [30]. We looked at several variables. Including variables that described the external environment of the class, which happened to be the Silicon Valley. In the next section we will describe some of the variables.

7. Towards a scientific understanding of design—how do you measure the design process?

Given that the work we have been doing spans close to 30 years, we will only provide a few highlights here to give a flavor of our research methodology and some of the results. In order to measure the design process as reliable as possible, we have opted for the use of video recordings for primary data. The video not only captures the behaviors of the designers, it also captures an aspect of the context of their work and interaction. The video can then be later analyzed to produce secondary data. In addi-

tion the subjects can be interviewed to provide their own interpretation in addition to the interpretation of the research team. Finally, as we developed new insights about the design process, we could review older tapes to re-analyze them from a different point of view. This method of analysis borrowed greatly from the field of Anthropology [31]. Coincidentally, Anthropology has been described as the most humanistic of the sciences and the most scientific of the humanities. This has served us well. In addition to Anthropology we have drawn from several other fields most notably engineering, computer science, and psychology. From engineering, we have borrowed metaphors for creating change in system. From computer science, we have designed interventions in form of tools to perturb the system. And from all four including anthropology, we have searched out frameworks to explain the behaviors we observed, which will now be presented.

7.1 Framework #1: Systematic reduction in size of the problem

Figure 2, illustrates the method we described earlier which allowed us to overcome the limitations of

scope of past approaches, while preserving the context. Thus the laboratory was set up to interact with the classroom which in turn was set up to interact with Industry.

7.2 Framework #2 Engineering feedback control

It can be said that one of the greatest contributions of engineering to the modern world is the feedback instrumentation system and methodology illustrated in Fig. 3. In it we measure and compare the actual performance of a system (the Plant) with the intended performance and make continuous adjustments to reduce the variance.

Using this analogy, we extended the feedback concept to include the process of design and the context of design. The resulting feedback system is shown in Fig. 4.

It is important to observe that this is not a study of the product, neither is it a study of the consumer. Instead it is a study of the producer. We were studying ourselves as engineering-designers. We were using our methods and techniques to improve ourselves. Next we will describe a useful framework for organizing the various measurements we worked with, followed by a sample list of the types

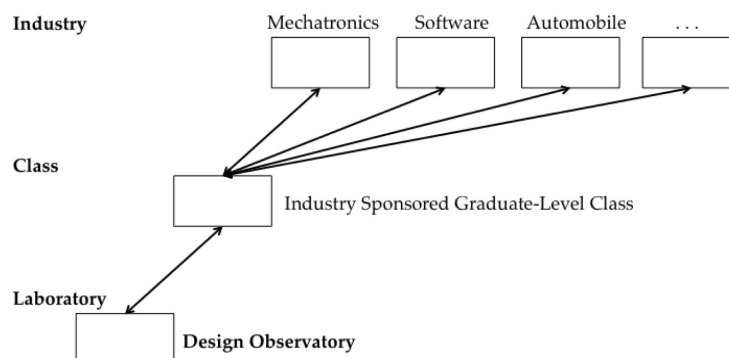


Fig. 2. Using an industry sponsored product-based learning class and a laboratory to overcome several limitations to developing a scientific understanding of design.

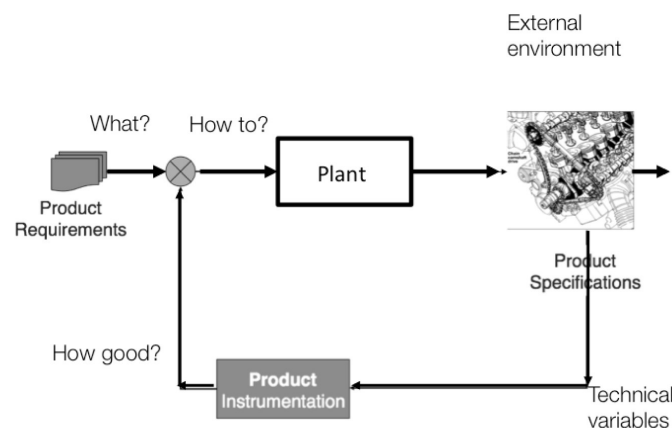


Fig. 3. A conceptual model of the basic feedback control system.

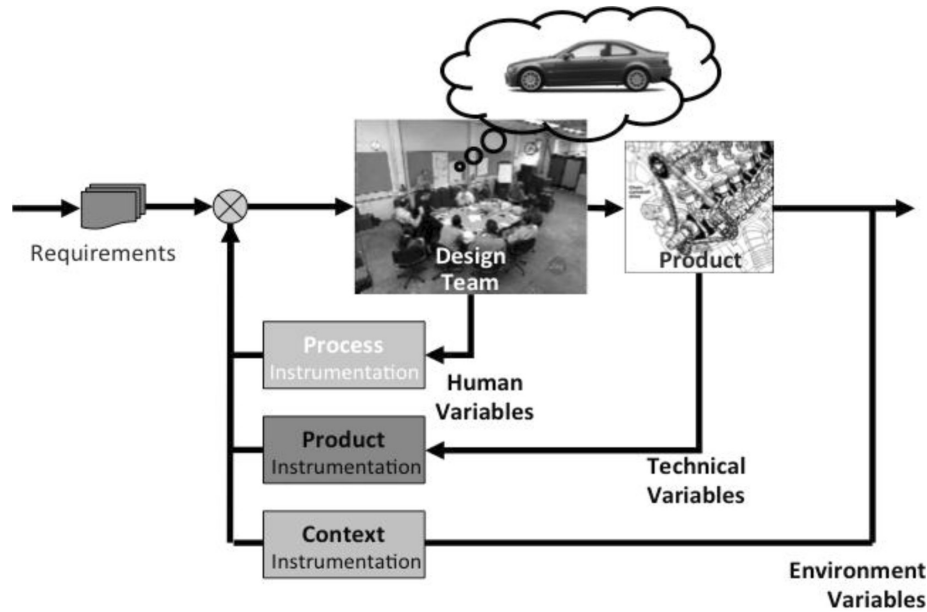


Fig. 4. A conceptual model of the basic feedback control system modified to study the design process in context.

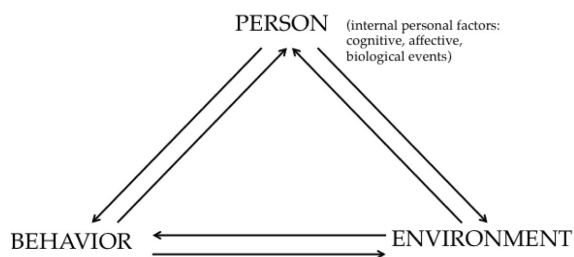


Fig. 5. The model of triadic reciprocal causation between a person's internal environment, their behavior, and their external environment.

of measurements we made from the video observations of engineering designers at work.

7.3 Framework #3 Model of triadic reciprocal causation

To give some coherence to the indicators and constructs we developed, we have found it useful to group them according to the model of triadic reciprocal causation [17].

According to this model, behavior, personal factors such as cognition, and environmental influences all operate as interlocking determinants that affect each other bi-directionally, See Fig. 5. In the following examples, primary data from our lab were investigated in the first two categories—Person and Behavior, which we describe as disposition and communication variables. The environmental variables, which we described as institution variables, have come from secondary data sources. Table 2 is a summary of all the variables we have collected and tested.

1. Brief description of generative design questions

In 2003, Ozgur Eris, found that the performance of design teams in creative problem solving was based on their ability to ask both of two types of questions [36]. The first, he called deep reasoning question. These are the types of question often taught in critical thinking classes. The second, he called generative design questions. These are questions that do not have to be true or false. He found that these sort of questions account for over 20 % of the questions creative teams pose during problem solving.

2. Brief description of noun phrases

In 1997, one of the authors found that the appearance of new noun phrases in design documentation could be used as a proxy for work progress during design [35]. Thus a rough count of unique noun phrases will give an early indicator of the novelty of the design.

3. Brief description of the emotion disposition

In early 2000, John Gottman published the result of work he had done in studying married couples. Gottman brought them into his lab, which was similar in several respects to our own lab. Based on 15 minutes of video based observation and analysis of the video, Gottman was able to predict which couples would divorce with an accuracy of 94% [47]. Malte Jung, then a graduate student at Stanford, applied Gottman's method to the study of design teams. Here the team was composed of software engineers engaged in an activity called pair programming. Jung was able to replicate Gott-

Table 2. A Sample of the Variables Measured in our lab grouped according to the model of triadic reciprocal causation

Communication Variables	
1	Gestures (Tang 1989) [32]
2	Ambiguity Preservation (Minneman 1991) [9]
3	Temporal Transitions (Baya, 1996) [33]
4	Categorical Transitions (Brereton 1997) [34]
5	Noun Phrase Generation (Mabogunje, 1997) [35]
6	Generative Design Questions (Eris, 2003) [36]
7	Describing Moment-to-Moment Concept Generation (Sonalkar 2012) [37]
Disposition Variables	
1	Team Emotional Balance (Jung, 2010) [38]
2	Team Composition (Schar, 2012) [39]
3	Team Diversity (Kress, 2012) [40]
Institution Variables	
1	Innovation and Economic Growth (Solow, 1956 & 1962) [41, 42]
2	Vision (Founding Narrative, ReVeL Handbook, 2005) [43]
3	Rules (Regulation of Capital Formation, Hwang & Horowitz, 2011) [44]
4	Network Density (Zoller, 2010) [45]
5	Age differences (Park et al, 2002) [46]

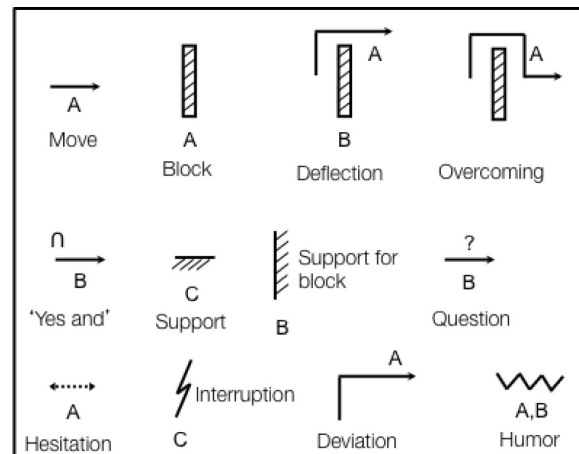
mans's study in two ways. First, he showed that teams with regulated affect i.e. the cumulative positive minus negative affect over time had a positive gradient, performed better than teams with unregulated affect i.e. the cumulative positive minus negative affect over time had a negative gradient [38].

4. Brief description of moment-to-moment concept generation

Improvisation is one of the most creative forms of interaction during design. Neeraj Sonalkar was convinced it held the key to understanding the sources of creative behavior amongst designers. However, he felt that the only way to demonstrate this was to track the interaction between designers on a moment-to-moment basis. Several attempts to find an appropriate representational schema failed. Finally using the rules of brainstorming and improvisational theatre as guides, he adapted a notational system called Force Dynamics Notation to represent the interaction sequences during design. From this he was able to develop twelve symbols for notating design interactions. See Fig. 6. From this he was able to demonstrate for the first time the conditions under which the exceptions to brainstorming rules and improvisation rules occur [37]. In addition, the visualization was able to shed light on how transitions between different types of information for example, proven facts (certainties) and plausible conjectures (possibilities), actually occur in design conversations.

5. Frustrations and paradigm shift

Despite the long list of variables and explanations we developed, we found most of the foregoing

**Fig. 6.** The 12 symbols of the Interaction Dynamic Notation System.

experiences most unsatisfactory. Like many other design researchers we entered the field of design research with backgrounds in analytical engineering and science—the “sciences of the natural” as Herbert Simon put it. But we were also designers at heart, wanting to change existing conditions for the better. Our dissatisfaction stemmed from the fact that on one hand we could do scientific research as observers detached from the situation; on the other hand we could do social science research with a contextual understanding but without changing the situation itself for the better; the growing list of variables and explanations made us despair. As we struggled with this problem we came upon an unusual source of inspiration in the person of Mozart. Mozart as recorded in the Amadeus Biography, made the claim that: “It is only through music, through opera that over 20 people can talk at

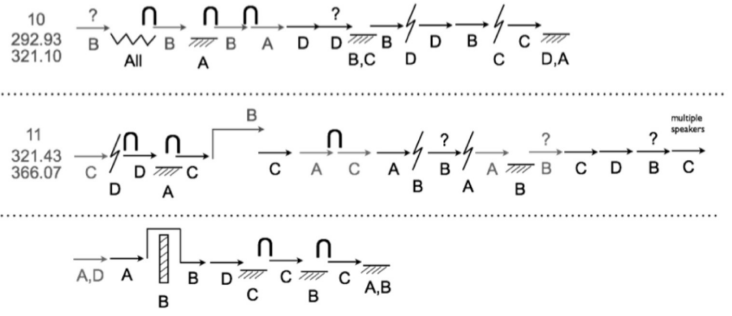
PHYSICS Mathematical equations The language of Mathematics has given us a rich way to describe patterns in nature	Continuity: $\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$ X - Momentum: $\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right]$ Y - Momentum: $\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} = -\frac{\partial p}{\partial y} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right]$ Z - Momentum: $\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho uw)}{\partial x} + \frac{\partial(\rho vw)}{\partial y} + \frac{\partial(\rho w^2)}{\partial z} = -\frac{\partial p}{\partial z} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right]$
CHEMISTRY Chemical equations Chemical equations and symbols have allowed us to depict the reversibility of chemical compositions	$\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Energy}$ 
BIOLOGY Genetic sequences This notation consists of 4 primitives – ATCG. These are the bases in DNA. A-Adenine, T-Thymine, C-Cytosine, G-Guanine.	KAP7 (AY791846) GAGAACTTCATTCTTCTGGGTAACCTGCTCTTCACATT KAP7 (X05638) ----- KAP7 (AY791846) CTATCCAAATCCTTCCCACCTCCTGCCACAATGACTCGTTT KAP7 (X05638) ----- KAP7 (AY791846) CTTTGTGCTGCGGAAGCTACTTCCCAGGCTATCCTTCCTAT KAP7 (X05638) ----- KAP7 (AY791846) GGAACCAATTTCCACAGGACCTTCAGAGCCACCCCTGA KAP7 (X05638) -----
DESIGN Interaction dynamics This notation consists of 12 basic primitives of generative human-human interaction. The sequence of these primitives is the DNA-equivalent of team-based Design.	

Fig. 8. Different ways of representing functions and behavior in Engineering including for the first time the behavior of the humans in the creative loop.

of innovation in semiconductor and software, and this will account for a good part of the economic growth.

8.2 Brief description of the significance of human vision in economic growth [4]

Earlier we quoted Frederick Terman's vision of the role a university could play in the life of a community. To better understand the significance of Terman's view it is useful to consider the basic circular flow of income in economics (see Fig. 9).

In this figure, households supply the factors of production to a factor market. Business firms acquire their resources from this factor market, and sell their output for a profit in the product market, where households do their shopping. The income generated from the sales flows back to pay the rent for the land, wages for labor, and profit for

capital. One interpretation of Terman's vision is that universities themselves could be considered factor markets, where not only is knowledge traded but labor can be transformed fundamentally.

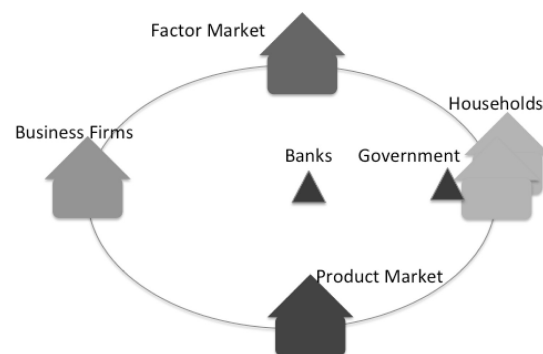


Fig. 9. The basic economic flow diagram.

In the older economic system, this would be equivalent to saying that universities can transform labor into landowners. However, we slipped into the knowledge economy and this is precisely what seemed to have happened in the Silicon Valley with the additional assistance of a new type of entrepreneurs called venture capitalists. The impact of venture capitalists on the Silicon Valley economy is profound and in the remaining of this section I will highlight three effects.

8.3 Brief Description of the significance of prevalence of dealmaker entrepreneurs and investors, with concurrent ties to multiple firms on rate of economic growth [45]

In 2010, Ted Zoller made an attempt to describe the relationship between dealmakers in 16 different geographical zones in the US—including Boston, Austin and the Silicon Valley. By deal makers, Zoller included angel investors, venture capitalists and others involved in the buying and selling of ventures, such as corporate investors responsible for mergers and acquisitions. While Boston appeared to have a density comparable to the Silicon Valley, a deeper analysis showed that the activity of dealmakers in the Silicon Valley differed significantly from those in other regions. Specifically when he looked at the number of dealmakers who sat on the board of 10 or more companies, the Silicon Valley had about 90 while Boston and other regions of the country had a number in the neighborhood of 30 and below. When one thinks of venture capitalists as cross pollination agents facilitating recombination of ideas and ventures, a difference between 30 and 90 makes a big difference [45].

8.4 Brief description of the significance of age differences [46]

While network effects are social in nature, there is another difference, which is more biological in nature. One cannot help but notice the young ages of most founders in the Silicon Valley, and the older ages of most investors. It is also well known that most good investors bring more than money to the table. They bring their knowledge, experience, and network. In one curve the processing capacity

decreases with time, and in the other curve the world knowledge increases with time. In a study on aging by Park et al., they were able to show a general decrease in processing capacity with age, and a general increase in world knowledge with age [46]. In most regions of the world, the extra processing speed of your people is often discounted. In the Silicon Valley, it appears this has been taken advantage of. Thus a team of young founders and older investors has when compared to a computer, a large memory and a fast processor. This appears to be an optimal combination.

8.5 Brief description of the significance of change in behavior on economic growth [44]

Victor Hwang and Greg Horowitz are two venture capitalists that are connected with Innovation ecosystem. In 2012 they wrote a book titled—The Rainforest: Secrets to Building the Next Silicon Valley. What was important about their book was their distillation of the behavior of people in the Silicon Valley to a set of rules, which contrasted sharply with rules followed by entrepreneurs, investors, and other supporters elsewhere in the world. While most governments had followed the conventional wisdom of building industrial parks close to universities as a way to spark innovation in a geographical region—Hwang and Horowitz drew attention to the cognitive and affective changes required for the new knowledge economy. Notice the mentality of a land-based economy in the idea of industrial parks. We believe the proximity of industry to universities is important, but it is not the dominant factor. In Table 3 we list the differences Hwang and Horowitz drew between the rules of a production economy, and the rules of an innovation.

8.6 A comprehensive measurement system for design

When we took the foregoing into consideration, it seemed feasible that we were in fact describing two separate and distinct economic systems: One, type-1, powered by the traditional university and the other, type-A, powered by a non-traditional university like Stanford. This is represented in Fig. 10.

Table 3. The Rules of the Plantation (Industrial Production System), contrasted with the Rules of the Rainforest (Innovation Ecosystem) [44]

Rules of the Plantation	Rules of the Rainforest
Excel at your job	Break rules and dream
Be loyal to your team	Open doors and listen
Work with those you can depend on	Trust and be trusted
Seek a competitive edge	Seek fairness, not advantage
Do the job right the first time	Experiment and iterate together
Strive for perfection	Err, fail, and persist
Return favors	Pay it forward

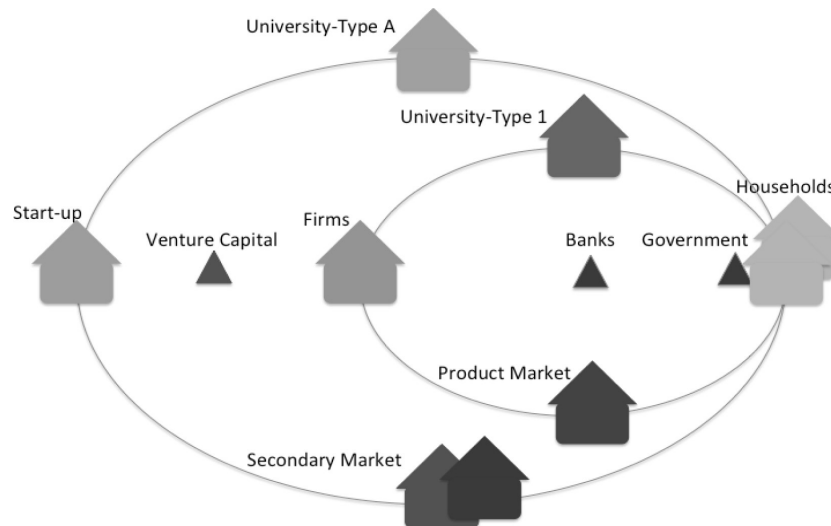


Fig. 10. It is hypothesized that the economy of the US may be modeled as two fairly independent circuits, whereby the outer dynamic circuit pulsates and helps to drive the larger inner circuit through innovation.

Given that start-ups are not always successful, it can be inferred that the outer loop is less permanent, and that it kind of pulsates with time. Having an economy such as the outer circle that pulsates in this way could have the effect of enabling the inner economy (the larger economy) to benefit and grow. In this way, the metaphor we made earlier, that the Silicon Valley was to the information revolution, what the steam engine was to the industrial revolution, could be realized.

9. Discussion part II—What are the academic implications?

By academic we refer to the practices of research and teaching of engineering design.

9.1 Consolidating the gains and advancing the research

On the research side, we now understand the cognitive and communicative basis of artifact creation and diffusion. We can measure artifact creation at the human interaction level, and can measure its diffusion at the economic level. We can also model the behaviors associated with artifact creation, and design new ones. We can create environments that impact a team's ability to develop innovative solutions to difficult problems. Design has been a difficult process to research and understand. We have been able to take advantage of several opportunities in the last 30 years to fit many pieces of design theories, methods, and activities together. There is still more work to be done, and we feel this phase of development will benefit greatly from a consolidation of the fundamentals of design as we have outlined in this paper.

9.2 Professionalizing design and advancing the teaching

We hope to have shown that context matters in design, and that by understanding and taking the context into account one can begin to derive a comprehensive measurement of design. It should therefore come as no surprise to readers that we see a strong need for design education to be run in a manner similar to a teaching hospital. The hands-on nature of design projects and the importance of context make this inevitable. Today's teaching hospitals consist of physicians who could be general practitioners or specialists, and then it has physicians who also have their doctoral degree. Finally it has a new class of researchers engaged in "translational research." These researchers are charged with translating breakthroughs in science to the bedside. At the beginning of this paper we mentioned that 24 incremental faculty billets had been approved by the Stanford provost for the bioengineering department. While we anticipate a similar expenditure in design, we expect the faculty roles to be more diverse—mathematics application, mechanics, and materials science research focused; design process research focused; teaching focused; and practice-focused. The first group is the current state of things in many schools with a few having the occasional professor of practice.

9.3 Potential impact

There are several areas of impact.

9.3.1 For the first time we are creating a feedback system for an innovation eco-system with Engineering Design Teams in the loop (See Fig. 4)

The expected impact is a program of continuous

improvement in Engineering education, production, innovation, and comprehension. Unlike ants and spiders who build ant hills and spider webs but cannot study themselves as they do it, we as engineers can now scientifically study ourselves as we engineer our products, services, ventures, ecosystems, and environment.

9.3.2 The second impact builds on the growing economic importance of design

There has been a shift in the relative economic value of commodities, products, services, and experiences over the last several years. We see the shift in pricing and in increasing order from commodity, to product, through service to the design of experiences [50, 51]. For example, according to Pine and Gilmore, it takes less than a half a dollar to produce a cup of coffee, and more than four dollars if it is bought or consumed in a high-end café such as Starbucks.

Similarly, it is now common knowledge that technology companies like Xerox make more money selling papers than actual machines. The design thinking movement while spearheading this trend will not only lead to early detection of un-met

user needs and market opportunities, it will also provide graduates of the professional school the sources of much coveted employment in the innovation industries.

9.3.3 The third impact is a much broader societal one

For example, there has been a steady decline in the number of fortune 500 companies created in Europe over the last 200 years. Furthermore, over the last 50 years, Europe has created only 3 new fortune 500 companies, while the US has created 57 [52]. It goes without saying that the Silicon Valley has been very influential in this area, and some credit goes to the vision of Stanford's School of Engineering, and their behavior in the context of those times.

The proposed professional school will be in a position to take the vision even further. We foresee a situation similar to Medical Education where the Flexner report of 1910 [53] ushered in the formal professionalization of medical education within the academy—using the John Hopkins University as the first model. Stanford will be ushering in through this professional school a strong triadic coupling between Mechanics, Mathematics Application, and

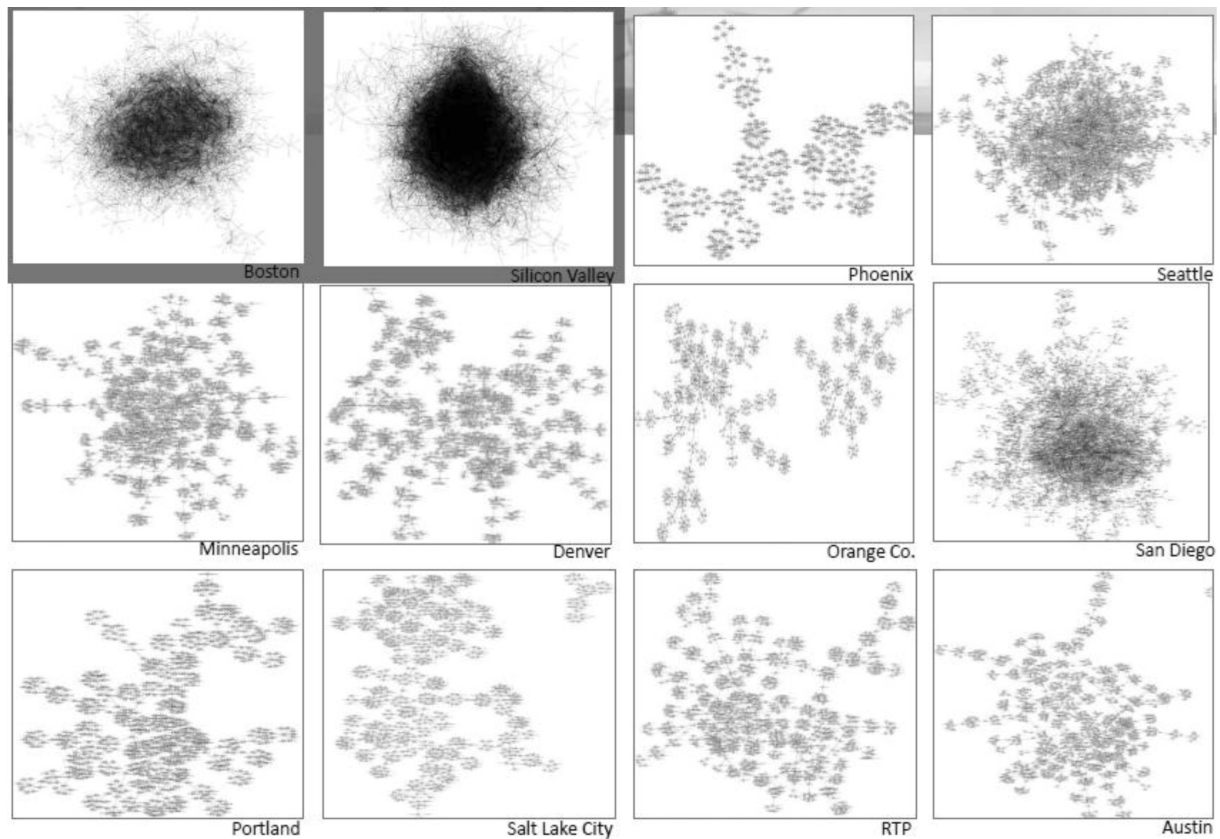


Fig. 11. The network of dealmakers (i.e. entrepreneurs and investors who combine characteristics of seriality, brokerage, and mediation) in 12 prominent technology metropolitan centers in the US. While the research described in this paper has been context specific, focusing on Stanford University, IDEO, the Silicon Valley Region, and specific people working in these environments, the control volume concept allows us to generalize the results to other geographies.

Materials Science research; Coaching, Teaching and Practice of Engineering-design; and Design Process Research focused on understanding the process of building the innovator and the innovative team in context.

In other words, while the research described has been context specific, the education enterprise provides a mechanism for corroboration and replication. We have focused on Stanford University, IDEO, the Silicon Valley Region, and specific people working in these environments as key elements within our control volume. The same control volume can be drawn in other locations to examine the pattern of interactions and the emergent consequences. Indeed such an attempt has been made in recent years. Ted Zoller, referenced earlier, in his research on technology based metropolitan areas in the United States, found that “the prevalence of dealmaker entrepreneurs and investors, with concurrent ties to multiple firms, is a better predictor of firm births than the prevalence and density of entrepreneurs and investors alone.” [45] Fig. 11, shows the entrepreneurial network analysis developed in his work.

10. Discussion part III—can design thinking really be a foundational science for engineering like physics, chemistry, biology, and mathematics?

In the course of doing this work and sharing the results, one of the interesting questions posed to us regards our claim that design thinking is a foundational science for Engineering. We believe it is a valid question, and one for which our response is an unequivocal yes. It is a valid question because it challenges both the questioner and us to re-examine our underlying beliefs about the nature and definition of engineering. Three such beliefs come to the fore here.

10.1 Within the engineering community, it is possible to distinguish two kinds of people

Those who came to the discipline because they were good at mathematics and those who love to build things. For the former, engineering is mathematics, and for the later, mathematics is a tool. This simplified grouping can be found in the other sciences where it can be mapped to the divide between the theorists and the experimentalists. Thus for a subset of the engineering community, the absence of a mathematical representation of design thinking could be troubling. For this group, we would like to point to the work of Armand Hatchuel and Benoit Weil at the Ecole de Mines in Paris, where they have developed a mathematical representation to explain the creative pro-

cess in design engineering [54]. For the other subset of the community, those who like to build things, the description of design thinking using an abstract representation does not feel concrete enough. For this group, the correspondence between the audio-visual records of the activity, the visual notation, and their felt sense of the experience has provided sufficient assurance of the objectivity of the notational scheme. As with most measurement systems, work needs to be done to continue to increase the accuracy, precision, reliability, and range of applicability.

10.2 The second underlying belief concerns a practice of modeling that is unique to engineering analysis though not ubiquitous

It is the control volume analysis. Given that it is not ubiquitous, and thus perhaps given little attention outside of the field of thermodynamics and fluid mechanics, we will provide an extended definition which has been taken from Walter Vincenti's well researched article on the history of the control volume analysis methodology [55].

“As used in thermodynamics and fluid mechanics, the control volume is an arbitrarily chosen volume, fixed in space and with fluid flowing through it. . . . An imaginary control surface, shown . . . by the closed dashed line surface. . . . fluid (either liquid or gas) must by definition flow through the control surface at some location, . . . This requirement stems from the fact that the control volume is designed specifically to analyze problems involving the flow of fluids, as occurs in engines, rockets, boilers, turbo-machinery, and a multitude of other devices with which engineers must deal. The importance of the control volume is that it provides, for flow devices, a convenient framework in which to apply the physical laws governing mass, momentum, energy, and (for an occasional problem) entropy. Such application leads to general integral equations involving changes inside the volume, transport of various quantities through the control surface, and forces acting on the material within the volume. These equations provide a prescription for doing the bookkeeping, so to speak, on the physical quantities going in and out of the control volume. The engineer chooses the volume in a given case both to facilitate application of the equations and with an eye toward the quantities he knows and those he needs to find. In the advice of one text, the boundaries should be put “either where you know something or where you want to know something.”

From the foregoing, it will be easy to see that the research presented here has involved a shift in the control volume. This in turn has led to a more comprehensive understanding of engineering. It should be self-evident that such a comprehensive view of engineering, which treats the engineer as a system variable, will lead to the creation of better artifacts. Fig. 12 shows the change in control volume from the traditional limited view of engi-

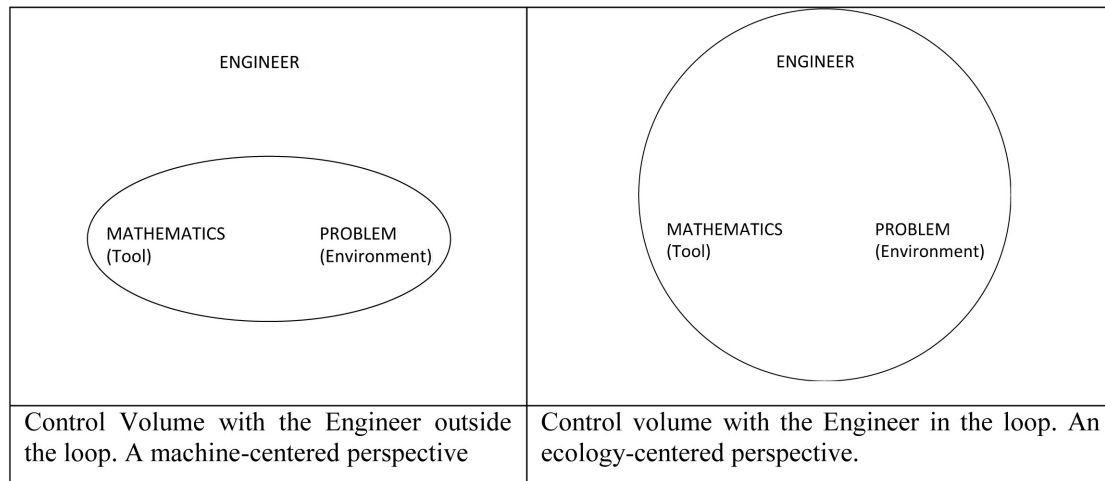


Fig. 12. Two views of engineering.

neering to a modern and more comprehensive one captured by the notation.

Vincenti concludes his description of the control volume as follows:

“Control-volume analysis thus comprises two elements: (1) the control volume (or control surface), and (2) the control-volume equations. With these elements the engineer has a systematic, explicit method for thinking about a large and important class of engineering devices.”

Correspondingly, we have considered (1) the Silicon Valley as the control volume, and (2) the interaction dynamic sequences as the control-volume equations.

As engineers, and engineering educators we quickly realize that in this new science, we can play two different roles with respect to our students. First, that of an independent scientific observer who transmits knowledge to a student, according to traditional means. Second, that of an engaged scientific experimenter, formulating new hypothesis based on the notation and the control volume, and testing these out in practice. We believe this second approach is more dynamic, relevant, innovative, and adaptive to a changing context of engineering education, industry, and the natural environment.

We have gone at length to quote Vincenti because as most will realize, Engineering History is not a required subject for most engineers today, hence the origin of some of our most useful methodologies is not well known.

10.3 The third underlying belief is perhaps the most daunting, and concerns the evolutionary nature of engineering and our relationship with mathematics

Mathematics in a way gives us a sense of control, and an instrument to make accurate predictions. To have a foundational science not based on mathe-

matics is new and threatening. Yet as humans and as scientists, we have an obligation to share and explain our results to the best of our knowledge and abilities. The notation is a culmination of over 25 years of research. In the introduction to an earlier work, we stated the following comments of a grammarian:

“In the newer technologies—notably in engineering—the {nomenclature} conventions are not systematic or clear; the {engineers} themselves are either unaware of the lack of clarity and system, or do not choose to make the effort to repair it. Therefore anyone who undertakes to read technical documents must make his way through agglomerations like these:

*the highest previously available intrinsic coercive force
single side band transmission
high frequency stability
high-energy particle accelerator
internal transducer excitation supply
the segmented multiple ablative chamber concept
combustion chamber crossover manifold coolant pas-
sages*

... This situation will stay with us until the {engineers} establish some firm conventions and hold to them as chemists and mathematicians hold to theirs.”

A Grammar of Standard English (Conner, 1968) [56]

It should be clear by now, that if engineers established some firm conventions and held on to them like chemists and mathematicians, scientific discovery and technological development in the last 200 years would have been very very slow. As engineers and scientists discover and build new tools, their language changes and evolves. Mathematical thinking, which is foundational to Engineering, and design thinking, which is also foundational to engineering appear to operate at different speeds and with different boundary constraints. Both are necessary foundations for the engineering disci-

pline, and for manipulating the impact of humans on the environment.

How can one provide a fundamental explanation of the structure and evolution of an anthill that is independent of the ants' individual and collective behaviors? This is a logic that makes design thinking foundational to engineering. Without it, the generative and divergent side of engineering remains in the dark, and light is only shed on the conservative, convergent side.

11. Conclusion

We began by showing that design thinking lies at the core of what engineers know and do. Next we discussed various approaches that have been used to improve the design thinking performance—from Allan Turing's early exploration of artificial intelligence to Engelbart et al's project on augmenting human intellect. We discussed several limitations in these approaches and how most of these could be overcome, save for the need to find standard ways to measure design process performance. Then, we showed that by taking into account the interaction pattern between designers and the context of their interaction, design thinking could indeed be measured in an integrative, comprehensive, and meaningfully effective manner. In addition, we demonstrated that by using the criteria for establishing the bioengineering department as precedence, we could conclude that design thinking is ready to become a new foundational science of engineering. Furthermore, taking into account the already immense impact of Turing, Terman, and Engelbart's work in improving design process performance in the absence of a measurement system, it stands to reason that we have barely scratched the surface of what is possible with the right industry-educational environment. We are, to quote one of our co-authors, in the early days of the Wright brothers and the Kitty Hawk.

References

1. S. J. Kline, *Conceptual foundations for multidisciplinary thinking*, Stanford University Press, Stanford, CA, 1995.
2. J. W. Carroll, *Laws of Nature*, Cambridge University Press, Cambridge, MA, 1994.
3. B. Flyvbjerg, *Making Social Science Matter: Why Social Inquiry Fails and How it can Succeed Again*, Cambridge University Press, Cambridge, MA, 2001.
4. S. W. Leslie and R. H. Kargon, Selling Silicon Valley: Frederick Terman's Model for Regional Advantage, *The Business History Review*, **70**(4), 1996, pp. 435–472.
5. Saxenian, *Regional Advantage: Culture and Competition in Silicon Valley and Route 128*, Harvard University Press, Boston, MA, 1994.
6. Solomon, *Design Thinking for Social Good—An Interview with David Kelley* <http://boingboing.net/2012/09/22/design-thinking-for-social-goo.html>, 2012.
7. L. Tischler, *Idea's David Kelley On "Design Thinking"* an Interview. <http://www.fastcompany.com/1139331/ideos-david-kelley-design-thinking>, 2009.
8. K. Turnali, *What Is Design Thinking?* Published on April 16, 2013 <http://blogs.sap.com/innovation/innovation/what-is-design-thinking-030869>, 2013.
9. S. Minneman, *The Social Construction of a Technical Reality: empirical studies of group engineering design practice*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 1991.
10. E. Wilson, *Consilience*. Alfred A. Knopf, NY, 1998.
11. H. A. Simon, *The Sciences of the Artificial*. Cambridge, MIT Press, MA, 1969.
12. D. C. Engelbart, *Augmenting Human Intellect: A Conceptual Framework*, SRI Summary Report AFOSR-3223, Prepared for: Director of Information Sciences, Air Force Office of Scientific Research, Washington DC, Contract AF 49(638)-1024, SRI Project No. 3578, 1962.
13. A.M. Turing, Computing machinery and intelligence, *MIND—A Quarterly Review of Psychology and Philosophy*, **59**(236), 1950, pp. 433–460.
14. Newell and H. A. Simon, *Human problem solving*, Prentice-Hall, Englewood Cliffs, NJ, 1972.
15. L. Vygotsky, *Thought and Language*, The M.I.T. Press, Boston, MA, 1934/1962.
16. J. Dewey, *Experience and Education*, Macmillan, Co, New York, NY, 1938.
17. Bandura, *Social Learning Theory*, General Learning Press, New York, NY, 1977.
18. L. Suchman, 1987, *Plans and Situated Actions: The Problem of Human/Machine Communication*, Cambridge University Press, New York, NY.
19. J. S. Brown, A. Collins and P. Duguid, Situated cognition and the culture of learning, *Educational Researcher*, **18**(1), 1989, pp. 32–42.
20. J. Lave and E. Wenger, *Situated Learning: Legitimate Peripheral Participation*, Cambridge University Press, New York, NY, 1990.
21. M. Donald, *Origins of the modern mind: Three stages in the evolution of culture and cognition*, Harvard University Press, Cambridge, MA, 1991.
22. W. J. Clancey, *Situated Cognition: On Human Knowledge and Computer Representations*, Cambridge University Press, New York, NY, 1997.
23. L. Leifer, *Suite-210: A Model For Global Product-Based-Learning With Corporate Partners*, Curriculum Innovation Awards: ASME Mechanical Engineering Departments Head Conference, San Diego, 1997.
24. L. E. C. Grinter, Report on Evaluation of Engineering Education, *American Society for Engineering Education*, Washington DC, 1995.
25. ABET, *Engineering Criteria 2000: Criteria for Accrediting Programs in Engineering in the United States*, The Accreditation Board for Engineering and Technology (ABET). Baltimore, MD, 1997.
26. J. C. Lucena, *Defending the Nation: US Policymaking in Science and Engineering Education from Sputnik to the War Against Terrorism*, University Press of America, Landham, MD, 2005.
27. M. J. Rabins, *Goals and Priorities for Research in Engineering Design*, ASME Press, New York, NY, 1986.
28. J. Womack, D. Jones and D. Roos, *The machine that changed the world: The Story of Lean Production*, Harper-Collins, New York, NY, 1991.
29. K. B. Clark and T. Fujimoto, *Product Development Performance: Strategy, Organization, and Management in the World Auto Industry*, Harvard Business School Press, Boston, MA, 1991.
30. L. J. Leifer, *Design Team Performance: Metrics and the impact of Technology. Evaluating Organizational Training*, S. M. Brown and C. Seidner, (Eds) Kluwer Academic Publishers, 1998.
31. B. Jordan and A. Henderson, Interaction Analysis: Foundations and Practice, *The Journal of the Learning Sciences* **4**(1), 1995, pp. 39–103.
32. J. Tang, *Toward an Understanding of the Use of Shared Workspaces by Design Teams*, Doctoral Dissertation,

- Mechanical Engineering, Stanford University, Stanford CA, 1989.
33. V. Baya, *Information Handling Behavior of Engineers in Conceptual Design: Three Experiments*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 1996.
 34. M. Brereton, *The Role of Hardware in Learning Engineering Fundamentals: An Empirical Study of Engineering Design and Product Analysis Activity*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 1999.
 35. A. Mabogunje, *Measuring Conceptual Design Performance in Mechanical Engineering: A Question Based Approach*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 1997.
 36. O. Eris, *Perceiving, comprehending and measuring design activity through the questions asked while designing*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 2002.
 37. N. S. Sonalkar, *A visual representation to characterize moment-to-moment concept generation through interpersonal interactions in engineering design teams*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 2012.
 38. M. F. Jung, *Engineering team performance and emotion: affective interaction dynamics as indicators of design team performance*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 2010.
 39. M. F. Schar, *Pivot thinking and the differential sharing of information within new product development teams*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 2011.
 40. G. Kress, *The Effects of Team Member Intrinsic Differences on Emergent Team Dynamics and Long-Term Innovative Performance in Engineering Design Teams*. Doctoral Dissertation, Mechanical Engineering, Stanford University, Stanford CA, 2012.
 41. R. M. Solow, A contribution to the theory of economic growth, *Quarterly Journal of Economics*, **70**(1), 1956, pp. 65–94.
 42. R. M. Solow, Technological Progress, Capital Formation and Economic Growth, *American Economic Review*, **52**, 1962, pp. 76–86.
 43. Realtime Venture design Lab, *ReVeL Research Primer*, Internal Report, Kozmetsky Global Collaboratory, Stanford, CA, 2005.
 44. V. W. Hwang and G. Horowitz, *The Rainforest: The Secret to Building the Next Silicon Valley*, Regenwald, Los Altos Hills, California, 2012.
 45. T. D. Zoller, *The Dealmaker Milieu: The Anatomy of Social Capital in Entrepreneurial Economies*. Doctoral Dissertation, University of North Carolina at Chapel Hill, NC, 2010.
 46. D. C. Park, G. Lautenschlager, T. Hedden, N. Davidson, A. D. Smith and P. Smith, Models of visuospatial and verbal memory across the adult life span, *Psychology and Aging*, **17**(2), 2002, pp. 299–320.
 47. J. M. Gottman, *What Predicts Divorce? The Relationship Between Marital Processes and Marital Outcomes*, Lawrence Erlbaum, Hillsdale NJ, 1994.
 48. Amadeus, Director—Milos Forman, 1984.
 49. N. Sonalkar, A. Mabogunje and L. Leifer, Developing a visual representation to characterize moment-to-moment concept generation in design teams, *International Journal of Design Creativity and Innovation*, **1**(2), 2013, pp. 93–108.
 50. A. Robertson and J. Woudhuysen, 4D Design: Applied Performance in the Experience Economy? presented at the *LIMIN Conference*, Brunel University, April 27–30, 2000.
 51. J. Pine and J. Gilmore, *The Experience Economy*, Harvard Business School Press, Boston, MA, 1999.
 52. The Economist, *European entrepreneurs, Les misérables*, 28th July 2012.
 53. A. Flexner, *Medical Education in the United States and Canada*, Science and Health Publications, Inc., Washington, DC, 1910.
 54. A. Hatchuel and B. Weil, A new approach of innovative design: an introduction to C-K theory, *International Conference on Engineering Design (ICED03)*, Stockholm, Sweden, 2003.
 55. W. G. Vincenti, Control-Volume Analysis: A Difference in Thinking between Engineering and Physics, *Technology and Culture*, **23**(2), 1982, pp. 145–174.
 56. J. E. Conner, *A Grammar of Standard English*, pp. 133–134, Boston: Houghton Mifflin, 1968.

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